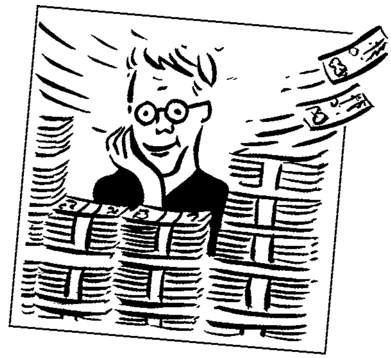
CHAPTER 22
value at risk



The aim of this Chapter...

... is to explain one common way of analyzing risk, to describe the basic methodology and also to hint at some of the difficulties associated with the methodology in practice. Many of the problems arise from the assumptions of stable parameters such as volatilities and correlations, and the absence of large jumps in the asset prices.

In this Chapter...

- the meaning of VaR
- how VaR is calculated in practice
- some of the difficulties associated with VaR for portfolios containing derivatives

22.1 INTRODUCTION

It is the mark of a prudent investor, be they a major bank with billions of dollars' worth of assets or a pensioner with just a few hundred, that they have some idea of the possible losses that may result from the typical movements of the financial markets. Having said that, there have been well-publicized examples where the institution had no idea what might result from some of their more exotic transactions, often involving derivatives.

As part of the search for more transparency in investments, there has grown up the concept of Value at Risk as a measure of the possible downside from an investment or portfolio.

22.2 DEFINITION OF VALUE AT RISK

One of the definitions of **Value at Risk** (VaR), and the definition now commonly intended, is the following.

> Value at Risk is an estimate,
> with a given degree of confidence,
> of how much one can lose from one's
> portfolio over a given time horizon.

The portfolio can be that of a single trader, with VaR measuring the risk that he is taking with the firm's money, or it can be the portfolio of the entire firm. The former measure will be of interest in calculating the trader's efficiency and the latter will be of interest to the owners of the firm who will want to know the likely effect of stock market movements on the bottom line.

The degree of confidence is typically set at 95%, 97.5%, 99%, etc. The time horizon of interest may be one day, say, for trading activities or months for portfolio management. It is supposed to be the timescale associated with the orderly liquidation of the portfolio, meaning the sale of assets at a sufficiently low rate for the sale to have little effect on the market. Thus the VaR is an estimate of a loss that can be realized, not just a 'paper' loss.

As an example of VaR, we may calculate (by the methods to be described here) that over the next week there is a 95% probability that we will lose no more than \$10m. We can write this as

$$\text{Prob}\,\{\delta V \leq -\$10\text{m}\} = 0.05,$$

where δV is the change in the portfolio's value. (I use $\delta\cdot$ for 'the change in' to emphasize that we are considering changes over a finite time.) In symbols,

$$\text{Prob}\,\{\delta V \leq -\text{VaR}\} = 1 - c,$$

where the degree of confidence is c, 95% in the above example.

VaR is calculated assuming normal market circumstances, meaning that extreme market conditions such as crashes are not considered, or are examined separately. Thus, effectively, VaR measures what can be expected to happen during the day-to-day operation of an institution.

The calculation of VaR requires at least having the following data: the current prices of all assets in the portfolio and their volatilities and the correlations between them. If the assets are traded we can take the prices from the market (**marking to market**). For OTC contracts we must use some 'approved' model for the prices, such as a Black–Scholes-type model; this is then **marking to model**. Usually, one assumes than the movement of the components of the portfolio are random and drawn from Normal distributions. I make that assumption here, but make a few general comments later on.

For more information about VaR and data sets for volatilities and correlations see `www.jpmorgan.com` and links therein.

22.3 VaR FOR A SINGLE ASSET

Let us begin by estimating the VaR for a portfolio consisting of a single asset.

We hold a quantity Δ of a stock with price S and volatility σ. We want to know with 99% certainty what is the maximum we can lose over the next week. I am deliberately using notation similar to that from the derivatives world, but there Δ would be the number held *short* in a hedged portfolio. Here Δ is the number held long.

In Figure 22.1 is the distribution of possible returns over the time horizon of one week. How do we calculate the VaR? First of all we are assuming that the distribution is Normal. Since the time horizon is so small, we can reasonably assume that the mean is zero. The standard deviation of the stock price over this time horizon is

$$\sigma S \left(\frac{1}{52}\right)^{1/2},$$

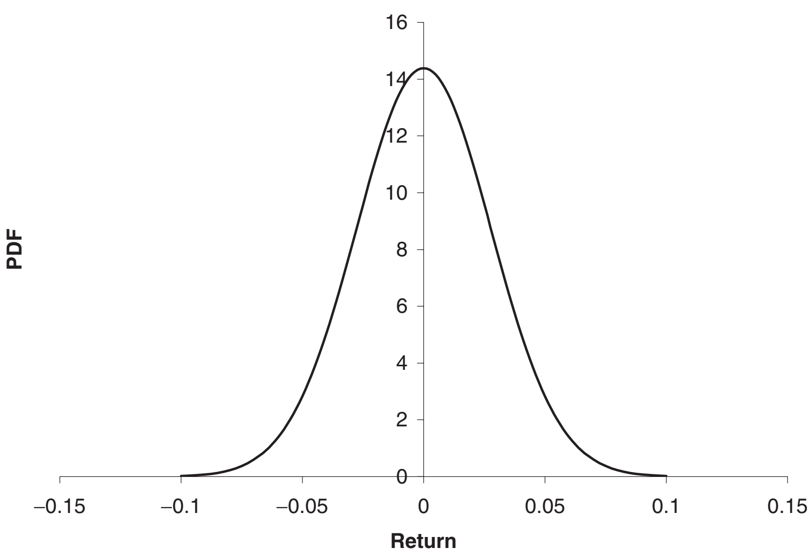


Figure 22.1 The distribution of future stock returns.

Table 22.1 Degree of confidence and the relationship with deviation from the mean.

Degree of confidence	Number of standard deviations from the mean
99%	2.326342
98%	2.053748
97%	1.88079
96%	1.750686
95%	1.644853
90%	1.281551

since the time step is $1/52$ of a year. Finally, we must calculate the position of the extreme left-hand tail of this distribution corresponding to $1\% = (100 - 99)\%$ of the events. We only need do this for the standardized Normal distribution because we can get to any other Normal distribution by scaling. Referring to Table 22.1, we see that the 99% confidence interval corresponds to 2.33 standard deviations from the mean. Since we hold a number Δ of the stock, the VaR is given by $2.33\sigma\,\Delta S(1/52)^{1/2}$.

Time Out...

Degree of confidence in Excel

How to get the number of standard deviations from the mean for a specified degree of confidence on a spreadsheet is shown below.

	A	B	C	D	E	F
1	**Mean**	**0.4**				
2	**Std dev**	**1.7**				
3						
4	**Conf.**	**No. SDs**	**Left-hand tail**			
5	99%	2.326342	-3.554781			
6	98%	2.053748	-3.091372			
7	97%	1.88079	-2.797342			
8	96%	1.750686	-2.576167			
9	95%	1.644853	-2.39625			
10	90%	1.281551	-1.778636			
11						
12						
13						
14						
15						

=B1-B8*B2

=NORMSINV(A10)

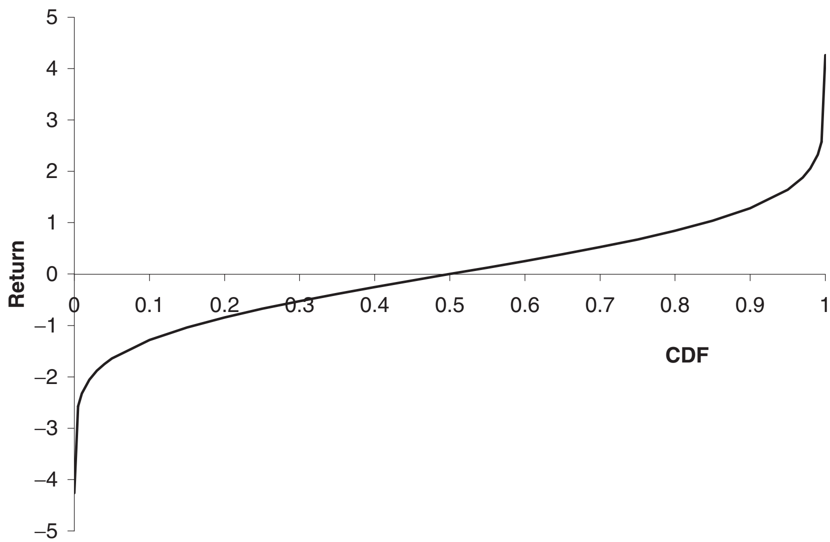


Figure 22.2 The inverse cumulative distribution function for the standardized Normal distribution.

More generally, if the time horizon is δt and the required degree of confidence is c, we have

$$\mathrm{VaR} = -\sigma\,\Delta S(\delta t)^{1/2}\alpha(1-c), \tag{22.1}$$

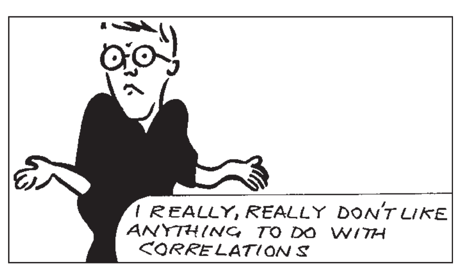
where $\alpha(\cdot)$ is the inverse cumulative distribution function for the standardized Normal distribution, shown in Figure 22.2.

In (22.1) we have assumed that the return on the asset is Normally distributed *with a mean of zero*. The assumption of zero mean is valid for short time horizons: the standard deviation of the return scales with the square root of time but the mean scales with time itself. For longer time horizons, the return is shifted to the right (one hopes) by an amount proportional to the time horizon. Thus for longer timescales, expression (22.1) should be modified to account for the drift of the asset value. If the rate of this drift is μ then (22.1) becomes

$$\mathrm{VaR} = \Delta S\left(\mu\ \delta t - \sigma\ \delta t^{1/2}\alpha(1-c)\right).$$

Note that I use the *real* drift rate and not the *risk-neutral*. I shall not worry about this adjustment for the rest of this chapter.

22.4 VaR FOR A PORTFOLIO



If we know the volatilities of all the assets in our portfolio and the correlations between them then we can calculate the VaR for the whole portfolio.

If the volatility of the ith asset is σ_i and the correlation between the ith and jth assets is ρ_{ij} (with $\rho_{ii} = 1$), then the VaR for the portfolio consisting

VaR for a portfolio on a spreadsheet

of M assets with a holding of Δ_i of the ith asset is

$$-\alpha(1-c)\delta t^{1/2}\sqrt{\sum_{j=1}^{M}\sum_{i=1}^{M}\Delta_i\Delta_j\sigma_i\sigma_j\rho_{ij}S_iS_j}.$$

Several obvious criticisms can be made of this definition of VaR: returns are not Normal, volatilities and correlations are notoriously difficult to measure, and it does not allow for derivatives in the portfolio. We discuss the first criticism later; I now describe in some detail ways of incorporating derivatives into the calculation.

22.5 VaR FOR DERIVATIVES

The key point about estimating VaR for a portfolio containing derivatives is that, even if the change in the underlying *is* Normal, the essential non-linearity in derivatives means that the change in the derivative can be far from Normal. Nevertheless, if we are concerned with very small movements in the underlying, for example over a very short time horizon, we may be able to approximate for the sensitivity of the portfolio to changes in the underlying by the option's delta. For larger movements we may need to take a higher-order approximation. We see these approaches and pitfalls next.

22.5.1 The delta approximation

Consider for a moment a portfolio of derivatives with a single underlying, S. The sensitivity of an option, or a portfolio of options, to the underlying is the delta, Δ. If the standard deviation of the distribution of the underlying is $\sigma S\ \delta t^{1/2}$ then the standard deviation of the distribution of the option position is

$$\sigma S\ \delta t^{1/2}\Delta.$$

Δ must here be the delta of the whole position, the sensitivity of all of the relevant options to the particular underlying, i.e. the sum of the deltas of all the option positions on the same underlying.

It is but a small, and obvious, step to the following estimate for the VaR of a portfolio containing options:

$$-\alpha(1-c)\delta t^{1/2}\sqrt{\sum_{j=1}^{M}\sum_{i=1}^{M}\Delta_i\Delta_j\sigma_i\sigma_j\rho_{ij}S_iS_j}.$$

Here Δ_i is the rate of change of the *portfolio* with respect to the ith asset.

22.5.2 The delta/gamma approximation

The delta approximation is satisfactory for small movements in the underlying. A better approximation may be achieved by going to higher order and incorporating the gamma or convexity effect.

I demonstrate this by example. Suppose that our portfolio consists of an option on a stock. The relationship between the change in the underlying, δS, and the change in the value of the option, δV, is

$$\delta V = \frac{\partial V}{\partial S}\delta S + \tfrac{1}{2}\frac{\partial^2 V}{\partial S^2}(\delta S)^2 + \frac{\partial V}{\partial t}\delta t + \cdots.$$

Since we are assuming that

$$\delta S = \mu S\ \delta t + \sigma S\ \delta t^{1/2}\phi,$$

where ϕ is drawn from a standardized Normal distribution, we can write

$$\delta V = \frac{\partial V}{\partial S}\sigma S\ \delta t^{1/2}\phi + \delta t\left(\frac{\partial V}{\partial S}\mu S + \tfrac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2\phi^2 + \frac{\partial V}{\partial t}\right) + \cdots.$$

This can be rewritten as

$$\delta V = \Delta\sigma S\ \delta t^{1/2}\phi + \delta t\left(\Delta\mu S + \tfrac{1}{2}\Gamma\sigma^2 S^2\phi^2 + \Theta\right) + \cdots. \qquad (22.2)$$

To leading order, the randomness in the option value is simply proportional to that in the underlying. To the next order there is a deterministic shift in δV due to the deterministic drift of S and the theta of the option. More importantly, however, the effect of the gamma is to introduce a term that is non-linear in the random component of δS.

In Figure 22.3 are shown three pictures. First, there is the assumed distribution for the change in the underlying. This is a Normal distribution with standard deviation $\sigma S\,\delta t^{1/2}$,

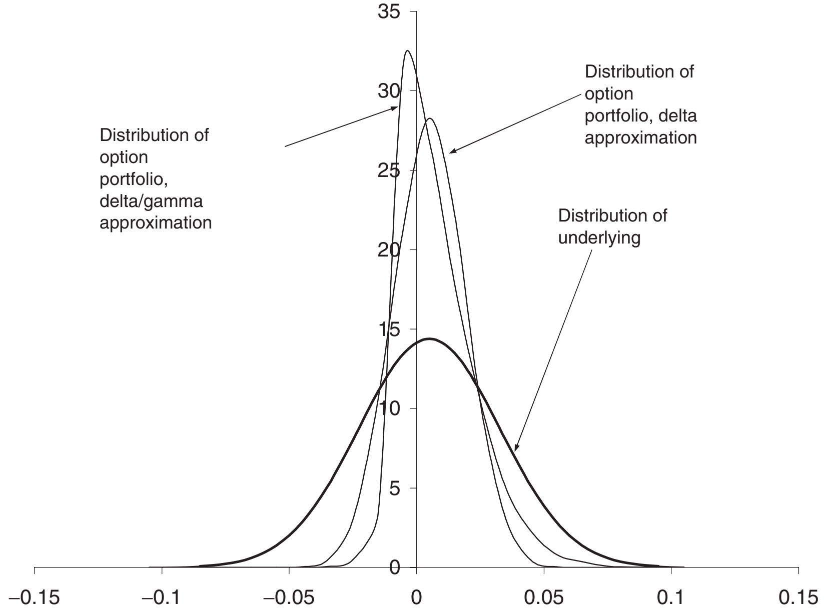


Figure 22.3 A Normal distribution for the change in the underlying (bold), the distribution for the change in the option assuming the delta approximation (another Normal distribution) and the distribution for the change in the option assuming the delta/gamma approximation (definitely not a Normal distribution).

drawn in bold in the figure. Second is shown the distribution for the change in the option assuming the delta approximation only. This is a Normal distribution with standard deviation $\Delta\sigma S\,\delta t^{1/2}$. Finally, there is the distribution for the change in the underlying assuming the delta/gamma approximation.

From this figure we can see that the distribution for the delta/gamma approximation is far from Normal. In fact, because expression (22.2) is quadratic in ϕ, δV must satisfy the following constraint

$$\delta V \geq -\frac{\Delta^2}{2\Gamma} \quad \text{if} \quad \Gamma > 0$$

or

$$\delta V \leq -\frac{\Delta^2}{2\Gamma} \quad \text{if} \quad \Gamma < 0.$$

The extreme value is attained when

$$\phi = -\frac{\Delta}{\sigma S\Gamma\,\delta t^{1/2}}.$$

The question to ask is then 'Is this critical value for ϕ in the part of the tail in which we are interested?' If it is not then the delta approximation may be satisfactory, otherwise it will not be. If we cannot use an approximation we may have to run simulations using valuation formulæ.

One obvious conclusion to be drawn is that positive gamma is good for a portfolio and negative gamma is bad. With a positive gamma the downside is limited, but with a negative gamma it is the upside that is limited.

22.5.3 Use of valuation models

The obvious way around the problems associated with non-linear instruments is to use a simulation for the random behavior of the underlyings and then use valuation formulæ or algorithms to deduce the distribution of the changes in the whole portfolio. This is the ultimate solution to the problem but has the disadvantage that it can be very slow. After all, we may want to run tens of thousands of simulations but if we must solve a multifactor partial differential equation each time then we find that it will take far too long to calculate the VaR.

22.5.4 Fixed-income portfolios

When the asset or portfolio has interest rate dependence then it is usual to treat the yield to maturity on each instrument as the Normally distributed variable. Yields on different instruments are then suitably correlated. The relationship of price to change in yield is via duration (and convexity at higher order). So our fixed-income asset can be thought of as a derivative of the yield. The VaR is then estimated using duration in place of delta (and convexity in place of gamma) in the obvious way.

22.6 SIMULATIONS

Two simulation methods used in VaR estimation are **Monte Carlo**, based on the generation of Normally distributed random numbers, and **bootstrapping** using actual asset price movements taken from historical data.

Within these two simulation methods, there are two ways to generate future scenarios, depending on the timescale of interest and the timescale for one's model or data. If one is interested in a horizon of one year and one has a model or data for returns with this same horizon, then this is easily used to generate a distribution of future scenarios. On the other hand, if the model or data are for a shorter timescale, a stochastic differential equation or daily data, say, and the horizon is one year, then the model must be used to build up a one-year distribution by generating whole year-long paths of the asset. This is more time consuming but is important for path-dependent contracts when the whole path taken must obviously be modeled.

Remember, the simulation must use *real* returns and not *risk-neutral*.

22.6.1 Monte Carlo

Monte Carlo simulation is the generation of a distribution of returns and/or asset price paths by the use of random numbers. This subject is discussed in great depth in Chapter 29. The technique can be applied to VaR: using numbers, ϕ, drawn from a Normal distribution, to build up a distribution of future scenarios. For each of these scenarios use some pricing methodology to calculate the value of a portfolio (of the underlying asset and its options) and thus directly estimate its VaR.

22.6.2 Bootstrapping

Another method for generating a series of random movements in assets is to use historical data. Again, there are two possible ways of generating future scenarios: a one-step procedure if you have a model for the distribution of returns over the required time horizon, or a multi-step procedure if you only have data/model for short periods and want to model a longer time horizon.

The data that we use will consist of daily returns, say, for all underlying assets going back several years. The data for each day is recorded as a vector, with one entry per asset.

Suppose we have real time-series data for N assets and that our data is daily data stretching back four years, resulting in approximately 1000 daily *returns* for each asset. We are going to use these returns for simulation purposes. This is done as follows.

Assign an 'index' to each daily change. That is, we assign 1000 numbers, one for each vector of returns. To visualize this, imagine writing the returns for all of the N assets on the blank pages of a notebook. On page 1 we write the returns in asset values that occurred from 4th August 2000 to 5th August 2000. On page 2 we do the same, but for the returns from 5th August to 6th August 2000. On page 3... from 6th to 7th August, etc. We will fill 1000 pages if we have 1000 data sets. Now, draw a number from 1 to 1000, uniformly distributed; it is 534. Go to page 534 in the notebook. Change all assets from today's value by the vector of returns given on the page. Now draw another number between 1 and 1000 at random and repeat the process. Increment this new value again using one of the vectors.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Index**	**Prob.**	**TELEBRAS**	**ELETROBRAS**	**PETROBRAS**	**CVDR**	**USIMINAS**	**YPF**	**TAR**	**TEO**	**TGS**	**PEREZ**	**TELMEX**	**TELEVISA**
2	1	0.001	-0.0152210	0.0180185	0.0118345	-0.0240975	-0.0111733	0.0355909	0.0185764	0.0071943	0.0121214	-0.0182190	-0.0235305	-0.0153849
3	2	0.001	0.0091604	0.0072214	-0.0046130	0.0001039	-0.0226244	0.0207620	0.0121953	0.0106953	0.0000000	-0.0026393	0.0327898	0.0746615
4	3	0.001	-0.0498546	-0.0397883	-0.0324353	-0.0246926	-0.0115608	0.0068260	-0.0121953	-0.0106953	0.0119762	-0.0067506	-0.0139213	-0.0144930
5	4	0.001	-0.0357762	-0.0494712	-0.0358569	-0.1001323	-0.0602100	-0.0068260	-0.0123458	-0.0254097	0.0000000	-0.0144520	-0.0284379	-0.0371790
6	5	0.001	0.0033058	-0.0204013	0.0112413	0.0264184	0.0114388	-0.0068729	0.0061920	0.0036697	-0.0240976	0.0096238	0.0000000	0.0000000
7	6	0.001	0.0424306	0.0260111	-0.0255043	-0.0022909	0.0004155	0.0067340	0.0452054	0.0438519	0.0115608	0.0140863	0.0421608	0.0932575
8	7	0.001	0.0279317	0.0006216	0.0006216	0.0006216	-0.0234759	0.0040080	-0.0208341	0.0000000	-0.0114287	0.0000000	0.0123205	-0.0065147
9	8	0.001	-0.0139801	-0.0383626	-0.0346393	0.0161040	0.0122250	0.0000000	0.0259755	0.0176996	0.0114287	-0.0020299	0.0439192	0.0384663
10	9	0.001	-0.0052219	-0.0036563	0.0403377	0.0142520	0.0229896	0.0063898	0.0101524	0.0085349	0.0114287	-0.0019638	0.0000000	0.0171924
11	10	0.001	-0.0130693	0.0000000	-0.0116961	-0.0047282	0.0112996	0.0000000	0.0100503	0.0112677	0.0112996	0.0081304	-0.0267702	-0.0229895
12	11	0.001	-0.0106656	-0.0148639	0.0069306	-0.0048541	-0.0114030	0.0000000	-0.0100503	-0.0169496	-0.0112996	-0.0166294	-0.0116960	0.0057971
13	12	0.001	0.0079035	0.0184167	-0.0023392	-0.0240975	0.0112996	0.0000000	-0.0050633	-0.0085837	0.0335227	-0.0170073	0.0039139	0.0228581
14	13	0.001	0.0000000	-0.0072225	-0.0284021	0.0001035	-0.0226248	0.0000000	-0.0257084	-0.0380719	-0.0110498	-0.0063762	0.0000000	0.0112361
15	14	0.001	0.0000000	-0.0042230	0.0242792	-0.0004134	-0.0126835	-0.0061920	0.0100001	0.0057471	0.0109291	0.0144183	0.0411581	0.0745331
16	15	0.001	-0.0475377	-0.0138620	-0.0251059	-0.0444106	-0.0254146	-0.0062305	-0.0150379	-0.0057471	0.0000000	0.0300138	-0.0080972	0.0000000
17	16	0.001	0.0297428	0.0102652	0.0114445	0.0714950	-0.0008256	0.0415490	0.0277796	0.0309303	0.0408220	-0.0035537	0.0419109	0.0475023
18	17	0.001	-0.0037045	-0.0192060	-0.0248187	0.0078541	-0.0129914	-0.0116280	-0.0045977	0.0052771	0.0000000	0.0007187	-0.0075758	-0.0105264
19	18	0.001	-0.0208341	-0.0205823	-0.0264421	-0.0021947	-0.0267507	0.0231224	0.0091744	-0.0052771	-0.0304592	0.0152095	-0.0269766	-0.0268113
20	19	0.001	0.0568874	0.0828276	0.1052577	0.0477150	0.0612661	0.0271019	0.0753494	0.0746435	0.0388398	0.0599762	0.0498324	0.0438026
21	20	0.001	0.0655911	0.0209154	0.0209922	0.0420456	0.0224756	-0.0173917	0.0165293	0.0223058	0.0000000	0.0232360	0.0036036	0.0194181
22	21	0.001	-0.0091109	0.0309987	0.0615672	-0.0428678	0.0221646	0.0058309	0.0122201	0.0145988	0.0392207	0.0139353	-0.0592766	-0.0342332
23	22	0.001	0.0231470	-0.0036810	-0.0003084	0.0158210	0.0099482	0.0342891	0.0160646	0.0264121	0.0204089	0.0208262	0.0679507	0.0386522
24	23	0.001	0.0157998	-0.0014713	-0.0122804	-0.0094697	-0.0014375	0.0277026	0.0504962	0.0440396	0.0492710	0.0285456	-0.0073260	0.0280392
25	24	0.001	-0.0045147	-0.0054003	-0.0008614	-0.0020500	0.0379553	-0.0054795	-0.0191210	-0.0136988	0.0000000	0.0101176	-0.0073801	-0.0233111
26	25	0.001	0.0270287	0.0275012	0.0583711	0.0001022	0.0361421	0.0280130	-0.0038536	0.0069849	0.0000000	0.0034104	-0.0109690	-0.0432968
27	26	0.001	-0.0089286	0.0068362	0.0038686	0.0248968	-0.0087868	0.0000000	-0.0116506	-0.0093241	0.0000000	0.0110554	-0.0036832	-0.0044346
28	27	0.001	0.0133632	0.0232183	0.0512933	0.0202027	0.0436750	-0.0167135	-0.0117880	-0.0285055	-0.0099503	-0.0080555	0.0000000	0.0088496
29	28	0.001	-0.0110341	0.0417395	0.0176995	0.0506932	0.0085108	0.0000000	-0.0240012	-0.0268636	0.0000000	0.0005324	0.0000000	0.0043956
30	29	0.001	-0.0044544	-0.0289144	0.0069770	-0.0002423	-0.0002043	-0.0056023	0.0000000	0.0024907	0.0000000	-0.0037901	0.0000000	-0.0277796
31	30	0.001	0.0044544	0.0001022	-0.0344781	-0.0075315	0.0086859	0.0056023	0.0202027	0.0049628	0.0204089	0.0147792	0.0038241	0.0277796
32	31	0.001	-0.0044544	-0.0065956	-0.0225751	0.0151071	-0.0173439	0.0111112	-0.0040080	0.0024722	0.0396091	0.0090663	-0.0115164	-0.0184337
33	32	0.001	0.0088685	-0.0067409	0.0284793	0.0204063	0.0261791	0.0096270	0.0043197	-0.0052632	0.0000000	0.0184492	-0.0076923	0.1048796
34	33	0.001	-0.0178163	-0.0084174	0.0036982	0.0183391	-0.0186661	-0.0242436	-0.0350913	-0.0240332	-0.0202027	-0.0280518	-0.0234386	-0.1242977
35	34	0.001	-0.0205219	-0.0174839	-0.0199091	-0.0849882	-0.0120831	-0.0061539	-0.0134834	-0.0108697	0.0000000	-0.0075047	0.0039448	0.0097562
36	35	0.001	-0.0092167	0.0172806	0.0323986	0.0769500	-0.0145534	-0.0313505	-0.0368705	-0.0221003	0.0202027	-0.0096608	-0.0320027	-0.0600180
37	36	0.001	-0.0256977	-0.0470936	-0.0464976	-0.0325785	-0.0460766	-0.0128207	-0.0189579	-0.0340942	-0.0100503	-0.0547012	-0.0081633	0.0102565
38	37	0.001	0.0303167	0.0398480	0.0188954	0.0204067	0.0278547	0.0625204	0.0421608	0.0396653	0.0000000	0.0471675	0.0322609	0.0497615
39	38	0.001	0.0000000	0.0178185	0.0003050	0.0122735	0.0093548	0.0000000	0.0091325	0.0082988	0.0298530	0.0320480	0.0039604	-0.0097562
40	39	0.001	-0.0113079	-0.0069686	0.0628009	-0.0039604	0.0266682	0.0180185	-0.0091325	-0.0109690	0.0194181	0.0310527	-0.0039604	-0.0148151
41	40	0.001	-0.0479236	-0.0464835	-0.0264332	0.0000000	-0.0266682	-0.0180185	-0.0279088	-0.0198027	-0.0096619	-0.0065198	-0.0119762	-0.0150379
42	41	0.001	-0.0374621	-0.0690578	-0.0427517	-0.0537785	-0.0563828	0.0000000	-0.0094787	-0.0229895	-0.0196085	-0.0109291	-0.0161947	0.0050378

Figure 22.4 Spreadsheet showing bootstrap data.

Continue this process until the required time horizon has been reached. This is one realization of the path of the assets. Repeat this simulation to generate many, many possible realizations to get an accurate distribution of all future prices.

By this method we generate a distribution of possible future scenarios based on historical data.

Note how we keep together all asset changes that happen on a certain date. By doing this we ensure that we capture any correlation that there may be between assets.

This method of bootstrapping is very simple to implement. The advantages of this method are that it naturally incorporates any correlation between assets, and any non-Normality in asset price changes. It does not capture any autocorrelation in the data, but then neither does a Monte Carlo simulation in its basic form. The main disadvantage is that it requires a lot of historical data that may correspond to completely different economic circumstances than those that currently apply.

In Figure 22.4 is shown the daily historical returns for several stocks and the 'index' used in the random choice.

22.7 USE OF VaR AS A PERFORMANCE MEASURE

One of the uses of VaR is in the measurement of performance of banks, desks or individual traders. In the past, 'trading talent' has been measured purely in terms of profit; a trader's bonus is related to that profit. This encourages traders to take risks; think of tossing a coin with you receiving a percentage of the profit but without the downside (which is taken by the bank), how much would you bet? A better measure of trading talent might take into account the risk in such a bet, and reward a good return-to-risk ratio. The ratio

$$\frac{\text{return in excess of risk-free}}{\text{volatility}} = \frac{\mu - r}{\sigma},$$

the **Sharpe ratio**, is such a measure. Alternatively, use VaR as the measure of risk and profit/loss as the measure of return:

$$\frac{\text{daily P\&L}}{\text{daily VaR}}.$$

22.8 INTRODUCTORY EXTREME VALUE THEORY

More modern techniques for estimating tail risk use **Extreme Value Theory**. The idea is to more accurately represent the outer limits of returns distributions since this is where the most important risk is. Throw Normal distributions away, their tails are far too thin to capture the frequent market crashes (and rallies). I won't go into the details here, a whole book could be written on this subject (and has been, Embrechts *et al*., 1997, a very good book).

22.8.1 Some EVT results

Distribution of maxima/minima: If X_i are independent, identically distributed random variables and

$$x = \max(X_1, X_2, \ldots, X_n)$$

then the distribution of x converges to

$$\exp\left(-\left(1 + \frac{\xi(x-\mu)}{\sigma}\right)^{-1/\xi}\right).$$

When $\xi = 0$ this is a Gumbel distribution, $\xi < 0$ a Weibull and $\xi > 0$ Frechet. Frechet is the one of interest in finance because it is associated with fat tails.

Peaks Over Threshold: Consider the probability that loss exceeds u by an amount y (given that the threshold *u has* been exceeded):

$$F_u(y) = P(X - u \leq y | X > u).$$

This can be approximated by a Generalized Pareto Distribution:

$$1 - \left(1 + \frac{\xi X}{\beta}\right)^{-1/\xi}.$$

For heavy tails we have $\xi > 0$ in which case not all moments exist:

$$E[X^k] = \infty \quad \text{for } k \geq 1/\xi.$$

The parameters in the models are fitted by Maximum Likelihood Estimation using historical data, for example, and from that we can extrapolate to the future.

Example: (From Alexander McNeil, 1998) Fit a Frechet distribution to the 28 annual maxima from 1960 to 16th October 1987, the business day before the big one. Now calculate probability of various returns. E.g. 50-year return level being the level which on average should only be exceeded in one year every 50 years. Result: 24%. And then the next day, 19th October : 20.4%. Note that in the data set the largest fall had been 'just' 6.7%.

22.9 COHERENCE

Some measures of risk make more sense than others. Artzner *et al*. (1997) have stated some properties that sensible risk measures ought to have, they call such sensible measures 'coherent.' If we use $\rho(X)$ to denote this risk measure for a set of outcomes X then the necessary properties are as follows:

1. Sub-additivity: $\rho(X + Y) \leq \rho(X) + \rho(Y)$. This just says that if you add two portfolios together the total risk can't get any worse than adding the two risks separately. Indeed, there may be cancelation effects or economies of scale that will make the risk better.
2. Monotonicity: If $X \leq Y$ for each scenario then $\rho(X) \leq \rho(Y)$. Pretty obviously, if one portfolio has better values than another under all scenarios then its risk will be better.
3. Positive homogeneity: For all $\lambda > 0$, $\rho(\lambda X) = \lambda\rho(X)$. Double your portfolio then you double your risk.
4. Translation invariance: For all constant c, $\rho(X + c) = \rho(X) - c$. Think of just adding cash to a portfolio.

Do the common measures of risk satisfy these reasonable criteria? Generally speaking, surprisingly and rather unfortunately not! For example, the classical VaR violates sub-additivity.

22.10 SUMMARY

The estimation of downside potential in any portfolio is clearly very important. Not having an idea of this could lead to the disappearance of a bank, and has. In practice, it is more important to the managers in banks, and not the traders. What do they care if their bank collapses as long as they can walk into a new job?

I have shown the simplest ways of estimating Value at Risk, but the subject gets much more complicated. 'Complicated' is not the right word; 'messy' and 'time-consuming' are better. And currently there are many software vendors, banks and academics touting their own versions in the hope of becoming the market standard. In Chapters 24 and 25 we'll see a few of these in more detail.

FURTHER READING

- See Lawrence (1996) for the application of the Value at Risk methodology to derivatives.
- See Chew (1996) for a wide ranging discussion of risk management issues and details of important real-life VaR 'crises.'

- See Jorion (1997) for further information about the mathematics of Value at Risk.
- The allocation of bank capital is addressed in Matten (1996).
- Alexander & Leigh (1997) and Alexander (1997a) discuss the estimation of covariance matrices for VaR.
- Artzner *et al*. (1997) discuss the properties that a sensible VaR model must possess.
- Lillo *et al*. (2002) introduce the concept of 'variety' as a measure of the dispersion of stocks.

EXERCISES

1. Assuming a Normal distribution, what percentage of returns are outside the negative two standard deviations from the mean? What is the mean of returns falling in this tail? (This is called the **censored mean**.)
2. What criticisms of Value at Risk as described here can you think of? Consider distributions other than Normal, discontinuous paths and non-linear instruments.